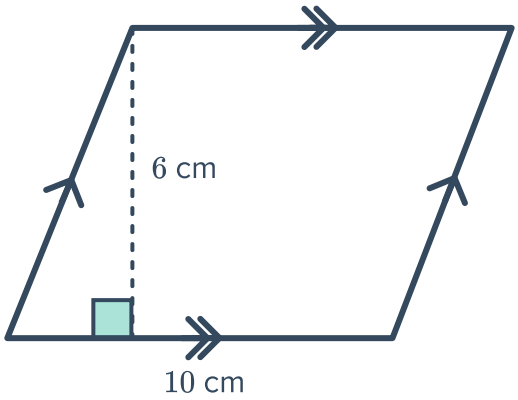
Parallelograms and rectangles

1 For the following parallelogram:



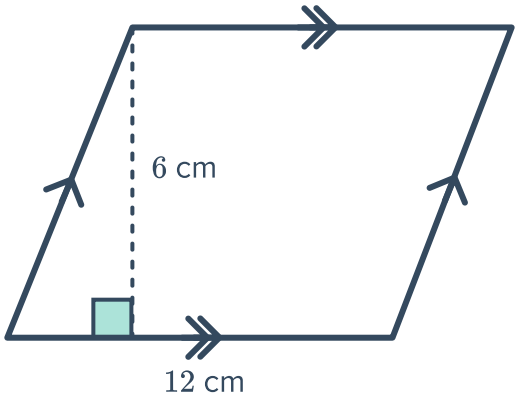
a If the parallelogram is formed into a rectangle, find:

i The length of the rectangle.

ii The width of the rectangle.

b Find the area of the parallelogram.

2 Find the area of the following parallelogram by rearranging it into a rectangle:



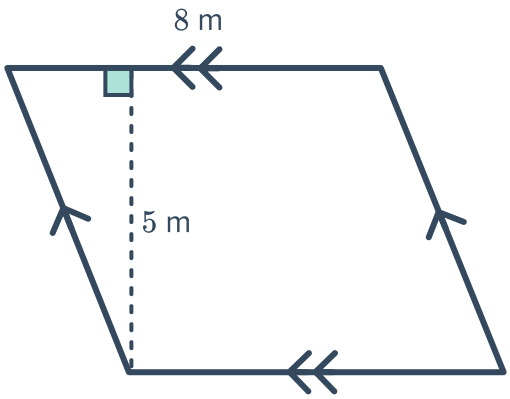
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Area of a parallelogram

3 Consider the following parallelogram:

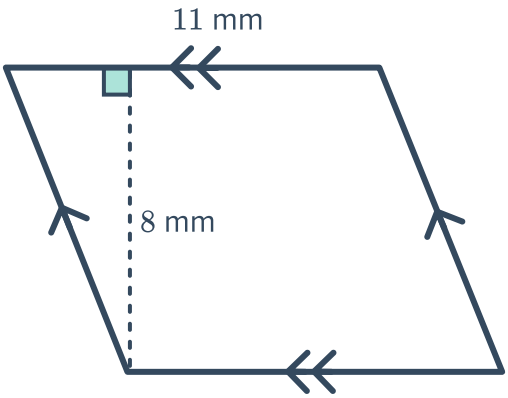
a Identify the base and the height.

b Find the area of the given parallelogram.

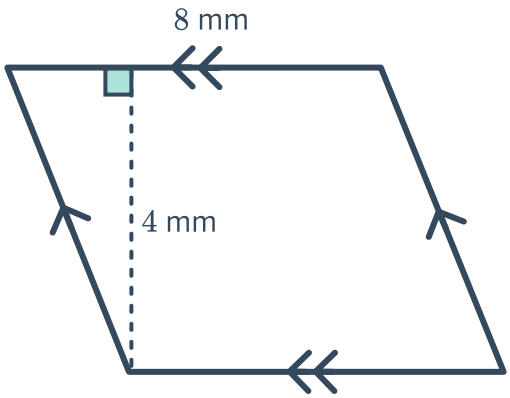


4 Find the area of the following parallelograms:

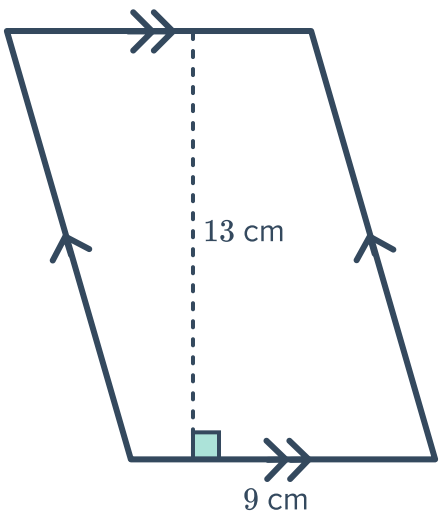
a



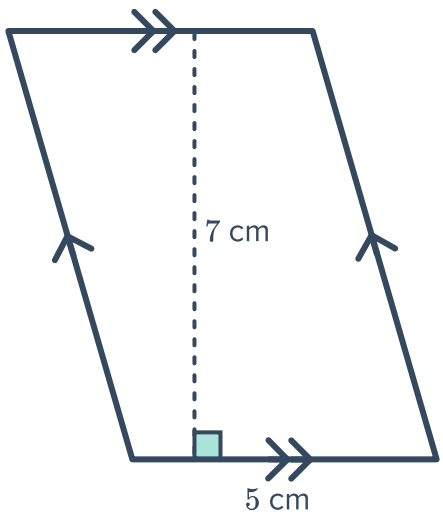
b



c

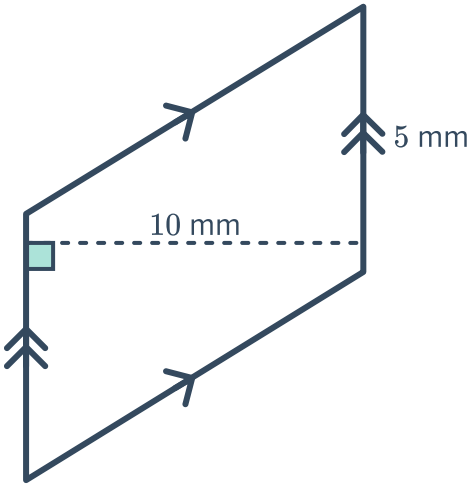


d

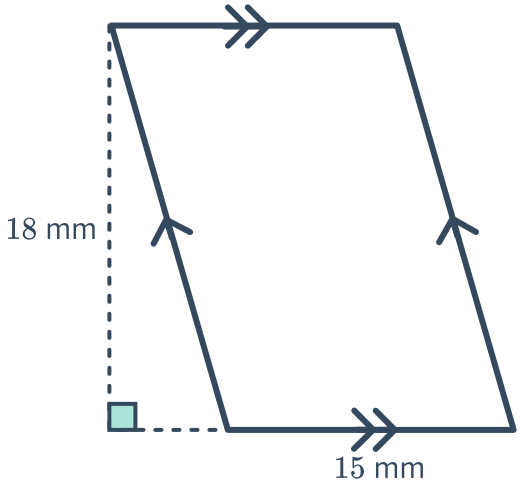


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

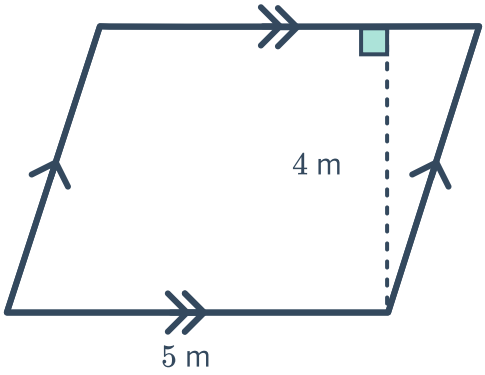
e



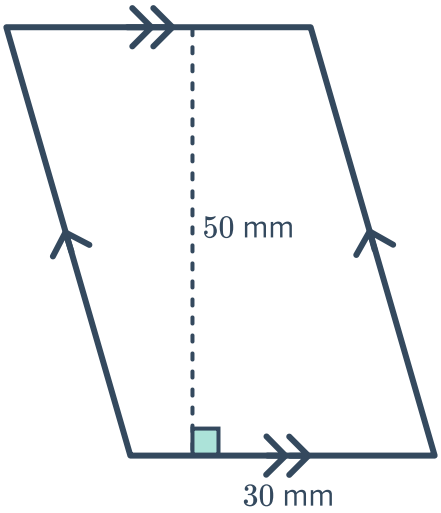
f



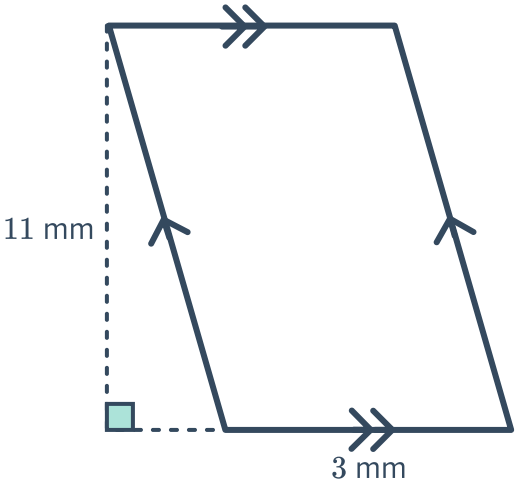
g



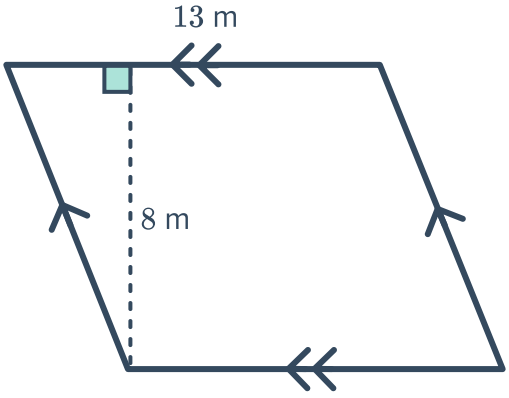
h



i

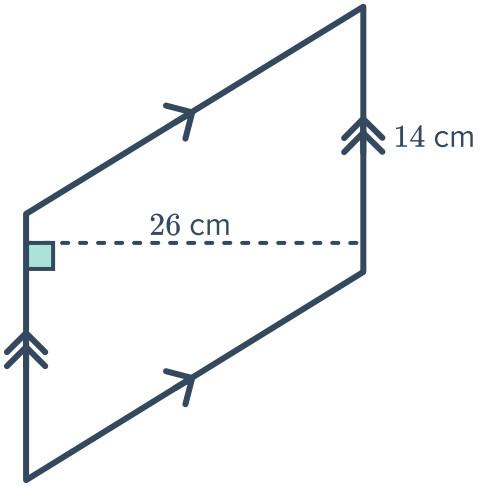


j

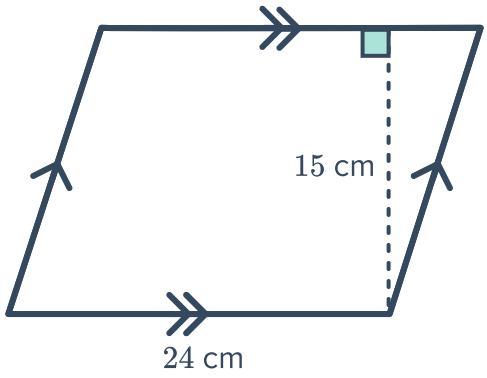


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

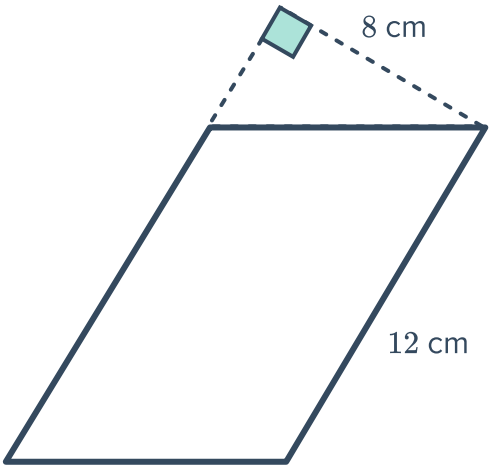
k



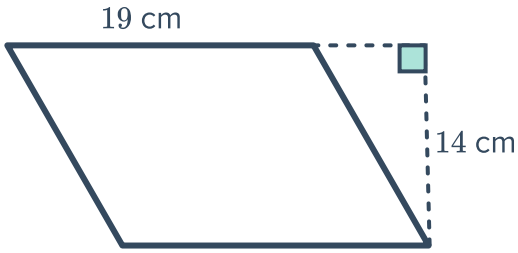
l



m



n



5 Find the area of the parallelograms with the following dimensions:

a The base is $5\ \text{mm}$, and the perpendicular height is $2\ \text{mm}$.

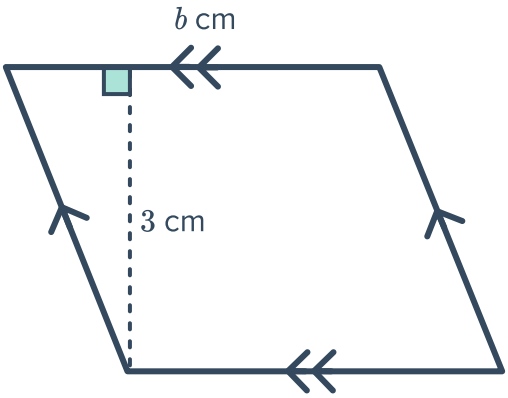
b The base is $15\ \text{cm}$ and the perpendicular height is $7\ \text{cm}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

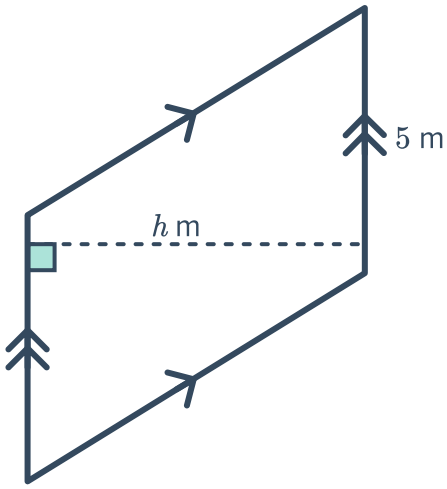
Dimensions from area

6 Find the value of b or h in the following parallelograms:

a Area $= 36\ \text{cm}^2$



b Area $= 35\ \text{m}^2$



7 Find the base length of a parallelogram whose area is $108\ \text{cm}^2$ and perpendicular height is $9\ \text{cm}$.

8 Find the perpendicular height of a parallelogram whose area is $99\ \text{m}^2$ and base is $9\ \text{m}$.

9 Determine whether the following could be the dimensions of a parallelogram with an area of $28\ \text{mm}^2$:

a Base $= 1\ \text{mm}$, height $= 28\ \text{mm}$

b Base $= 7\ \text{mm}$, height $= 4\ \text{mm}$

c Base $= 4\ \text{mm}$, height $= 7\ \text{mm}$

d Base $= 2\ \text{mm}$, height $= 28\ \text{mm}$

10 Complete the following table of base and height measurements for different parallelograms with an area of $60\ \text{cm}^2$:

Base (cm)		10	30
Height (cm)	12		
Area (cm^2)	60	60	60

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

11 Some car parks require the cars to park at an angle as shown. The dimensions of the car park are as given, where each individual parking space has a length of $5.1\ \text{m}$ and a width of $3.9\ \text{m}$. What area is needed to create an angled car park suitable for 7 cars?

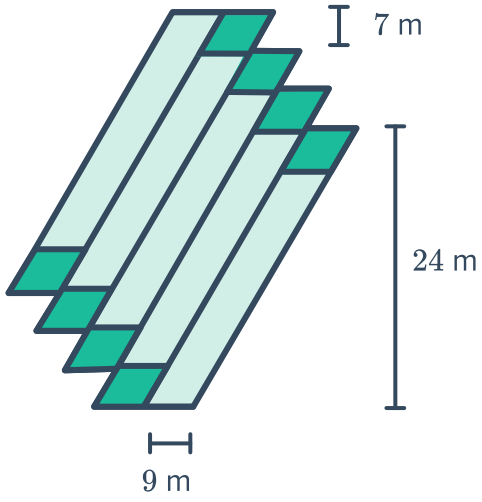


12 The tile pattern shown is made up of two different parallelograms in a tessellated pattern.

a Find the area covered by the smaller tiles.

b Find the area covered by the larger tiles.

c Calculate the total area covered.



13 James used some scrap paper to make a birthday card in the shape of a parallelogram. The base of the card is $17\ \text{cm}$ long, and the perpendicular height is $16\ \text{cm}$. Find the area of the card.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 A roof in the shape of a parallelogram is to be entirely covered by identical solar panels that are also in the shape of a parallelogram. Each solar panel measures $1.2\ \text{m}$ along the longest side and has a perpendicular height of $0.6\ \text{m}$.

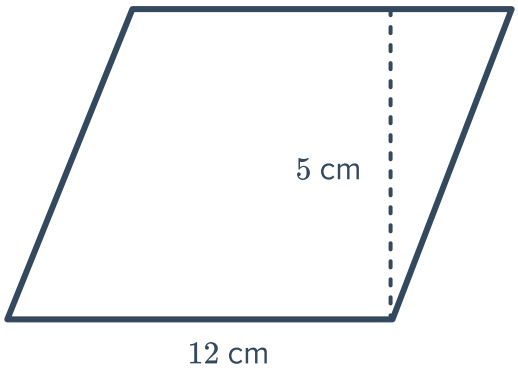
a Find the area covered by one solar panel.

b Find the number of solar panels to be installed if the roof measures $23.04\ \text{m}^2$ in area, and no part of the roof is to be left uncovered by solar panels.

15 An area measuring $2280\ \text{cm}^2$ is to be paved with identical tiles in the shape of parallelograms. Each tile measures $12\ \text{cm}$ along the base, and has a perpendicular height of $5\ \text{cm}$.

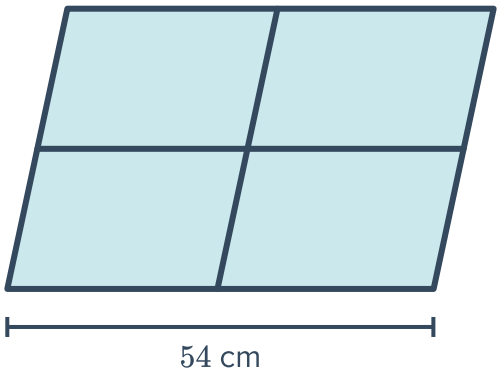
a Find the area that each tile covers.

b How many tiles are needed to cover the whole area?



16 A quilt is made by sewing together 4 identical parallelograms as shown in the following figure:

If the total area of the quilt is $1944\ \text{cm}^2$, calculate the perpendicular height of each parallelogram piece.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647